HWW3: multi-resolution HEALPix unstructured grids for WAVEWATCH III

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Abstract

We build unstructured WAVEWATCH III (WW3) grids whose nodes are the centres of HEALPix cells of different resolutions: fine near the coast, coarse in the open ocean. The model code is unchanged. Because every node is a NESTED HEALPix pixel, the model state is a multi-resolution HEALPix map. It can be stored as GRID4EARTH Zarr datasets and compared with satellite products on the same grid without interpolation. This note describes the grid construction, how it is validated, and a first North Atlantic simulation for January 2026. [TODO: results]

1 Motivation

Unstructured wave models usually rely on meshes from generic mesh generators. Their nodes do not belong to any global indexing system, so every comparison with observations or with another model needs an interpolation. HEALPix offers a hierarchical, equal-area and iso-latitude tessellation of the sphere. Each cell at N_{side} splits into four cells at $2N_{\mathrm{side}}$, so multi-resolution grids come from simple quadtree refinement, and each node has a global integer identifier (the NESTED *uniq* index). Placing the WW3 nodes at HEALPix cell centres keeps the flexibility of unstructured grids while making the outputs native to discrete global grid systems.

2 Grid construction

2.1 Refinement

We start from the cells at $N_{\mathrm{side}}^{\mathrm{min}}$ whose centres lie in the domain, expanded by a margin. A cell of angular radius r is split into its four children when its lower-bound distance to the coastline, $d - r$, is below the distance D_k of a refinement rule k whose target resolution $N_{\mathrm{side}}^{(k)}$ is finer than the cell. The coastline comes from the GSHHS land/ocean polylines (shapefile already available on Datarmor; the reader also accepts the native GSHHG binary format). It is densified to half the finest cell size and thinned to one point per cell at $4\,N_{\mathrm{side}}^{\mathrm{max}}$. Nearest-coast queries use buckets of HEALPix cells at $N_{\mathrm{side}}^{\mathrm{max}}/4$, searched by rings of neighbours. The number of levels is limited to $\log_2(N_{\mathrm{side}}^{\mathrm{max}}/N_{\mathrm{side}}^{\mathrm{min}}) \leq 4$.

2.2 2:1 balance

Leaves are then split until any two leaves sharing an edge *or a corner* differ by at most one level. A leaf at depth d is split when one of its eight same-depth neighbours is refined and one of that neighbour's children is refined as well.

2.3 Triangulation from the HEALPix adjacency

Every leaf registers the eight points of its boundary that it can share with a neighbour: its four corners and the midpoints of its four edges. The midpoints are the corners of its children. Identical

points coming from different leaves are merged by quantised unit vectors. Around a point p, the registered leaves sorted by angle in the tangent plane at p form a polygon of cell centres, the dual cell of p:

- 4 cells at a regular vertex;
- 3 cells at the eight base-face corners where only three cells meet;
- 3 cells at the midpoint of a coarse edge facing two fine cells.

Each maximal run of consecutive wet leaves with at least three members is triangulated by ear clipping, at each step clipping the ear with the largest minimum angle.

The mesh is conforming because two leaves sharing a boundary segment $[p, q]$ are consecutive around p and around q only. Their connecting edge therefore belongs to exactly the two polygons of p and q, and to at most one triangle in each. With the 2:1 balance, a quarter point of a coarse edge is registered by at most two fine leaves and never produces a triangle.

2.4 Ellipsoid

HEALPix is defined on a sphere. To use the WGS84 ellipsoid, we use the authalic sphere, which keeps HEALPix equal-area on the ellipsoid. The geodetic latitudes of all inputs (bathymetry, coastline, domain) are converted to authalic latitudes before hashing, with the exact formula. Node latitudes are converted back with a series in e^2 refined by two Newton steps; the round-trip error is below 10^{-12} rad. The difference between the two latitudes reaches 0.128° at 45°, i.e. more than 14 km, which is larger than the finest cells. A spherical option is kept for comparison.

2.5 Bathymetry and wet cells

Each GEBCO 2024 sample (15″) is assigned to the leaf containing it. A leaf is wet when at least half of its samples are wet and its mean wet depth exceeds 5 m. The node depth is the mean depth over the wet samples. Only the largest connected wet component is kept, which removes lakes and basins not connected to the ocean at the grid resolution.

3 Validation on synthetic cases

The mesh is checked automatically: no edge shared by more than two triangles, no interior edge traversed twice in the same direction, all triangles counter-clockwise, Euler characteristic $V - E + F = 1 - h$ for h holes, and the largest level jump across an edge. Table 1 gives the multi-resolution test: an island of radius 2° centred on a HEALPix base-face corner (45°E, 41.8°N), with $N_{\text{side}} = 32 \dots 512$ and rules (30 km, 512), (100 km, 256), (250 km, 128). Figure 1 shows the mesh around the island.

Quantity	Value
nodes (N_{side} 32/64/128/256/512)	94 / 194 / 231 / 297 / 494
triangles	2471
cells split by the 2:1 balance	29
non-manifold / inconsistent edges, clockwise triangles, pinch nodes	0 / 0 / 0 / 0
Euler characteristic (one island)	0
max level jump across an edge	1
min / max angle	17.0° / 143.7°
triangles with an angle $< 20°$	1

Table 1: Synthetic island test (`cargo test island_multiscale`).

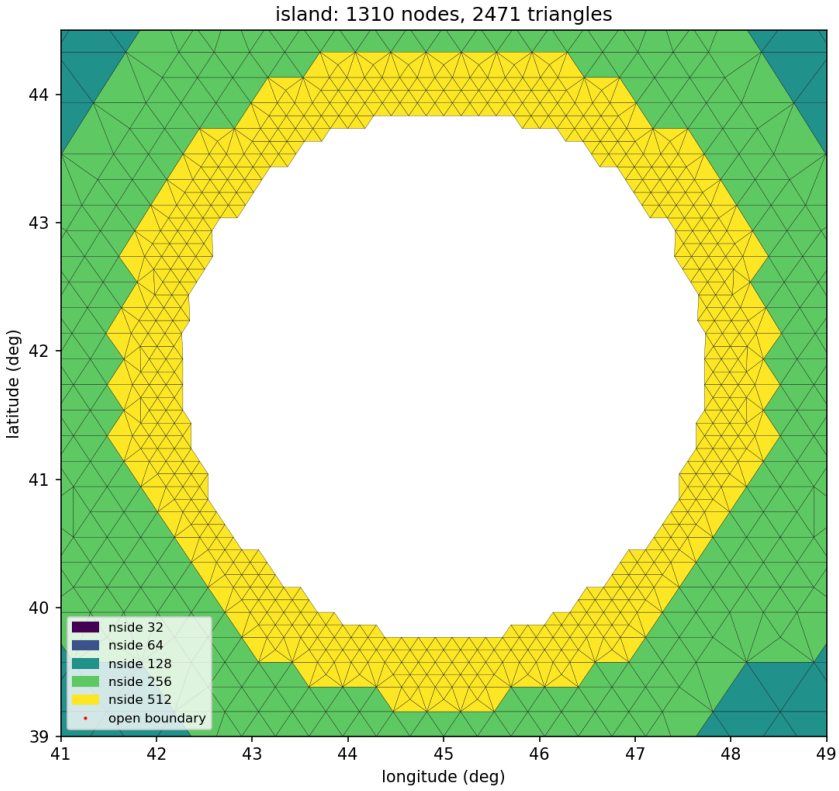


Figure 1: Multi-resolution HEALPix mesh around the synthetic island, coloured by N_{side}. The island sits on a base-face corner, where only three cells meet.

4 North Atlantic configuration

- Domain: 100°W–45°E, 0°–80°N, including the Mediterranean and the North Sea.
- Resolution: $N_{\mathrm{side}}^{\mathrm{min}} = 128$ (≈51 km) to $N_{\mathrm{side}}^{\mathrm{max}} = 2048$ (≈3.2 km).
- Refinement rules: 10 km → 2048, 50 km → 1024, 200 km → 512.
- Physics and numerics: WW3 `develop` (NOAA-EMC), ST4 source terms, 36 directions × 32 frequencies ($f_1 = 0.0373$ Hz, ratio 1.1), implicit unstructured scheme (PDLIB) with domain decomposition.
- Forcing (January 2026): ERA5 10 m wind; currents and sea level from Copernicus Marine; a fixed sea-ice obstruction taken from the January ERA5 sea-ice concentration.
- Open boundaries (southern boundary at the equator, eastern Barents Sea): spectra from an existing global run. [TODO: source of the spectra]

4.1 Grid statistics

The grid was built on Datarmor (PBS job 7893701, 8 cores) in 152 s wall clock (a first build, job 7893385, took 375 s with a peak resident memory of 5.8 GB); the GEBCO extraction (36 240 × 20 640 samples at 15 arcsec) takes 30 s. Coastline: GSHHS full-resolution land polylines, densified to 883 611 points. Table 2 summarises the mesh. The largest wet component is kept (902 components found); the Euler characteristic −693 corresponds to 694 islands.

Pinch nodes. WW3 walks the boundary around every node and stops (`w3triamd`: "dramatic inconsistency") when two triangle fans touch at a single node, a bow-tie that appears where water lies on two sides of a narrow spit. Such nodes pass the edge-based checks. The generator now keeps, at every such node, the largest fan and drops the triangles of the others, iterating to a fixed point: 671 triangles (0.07 %) were removed, and the validator counts pinch nodes (0 after the fix).

Open-boundary nodes (971) lie on the equator, at 80°N, on the 45°E cut of the Barents Sea and on the 100°W cut of the Canadian archipelago (Figure 2).

Quantity	Valu
nodes (N_{side} 128/256/512/1024/2048)	10 160 / 6 274 / 73 088 / 135 561 / 277 60
nodes, total	502 69
triangles	949 19
leaves of the quadtree (wet)	1 168 555 (573 61
cells split by the 2:1 balance	2 45
non-manifold / inconsistent edges, clockwise triangles, pinch nodes	0 / 0 / 0 /
Euler characteristic	−69
max level jump across an edge	
min / max angle	17.5° / 139.
triangles with an angle $< 20°$ / $> 120°$	3 781 / 3 07
open-boundary nodes	97
time: refinement + balance / cell bathymetry / triangulation	70 / 23 / 11

Table 2: North Atlantic grid (`configs/natl_datarmor.json`).

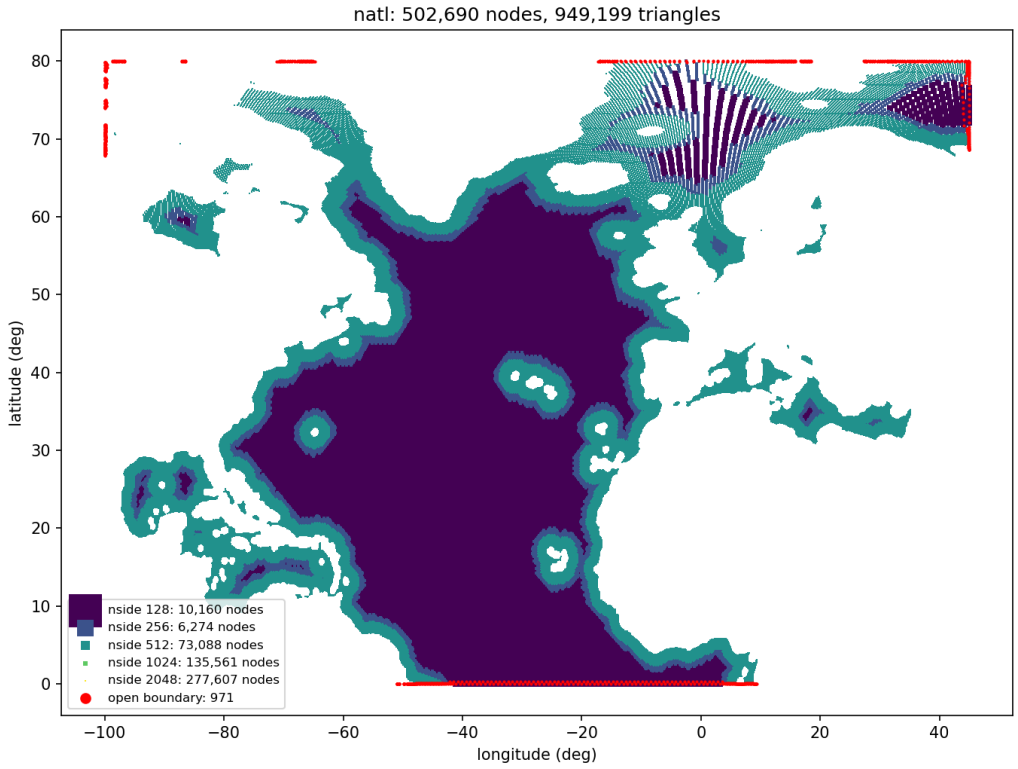


Figure 2: North Atlantic grid nodes coloured by N_{side}; open-boundary nodes in red.

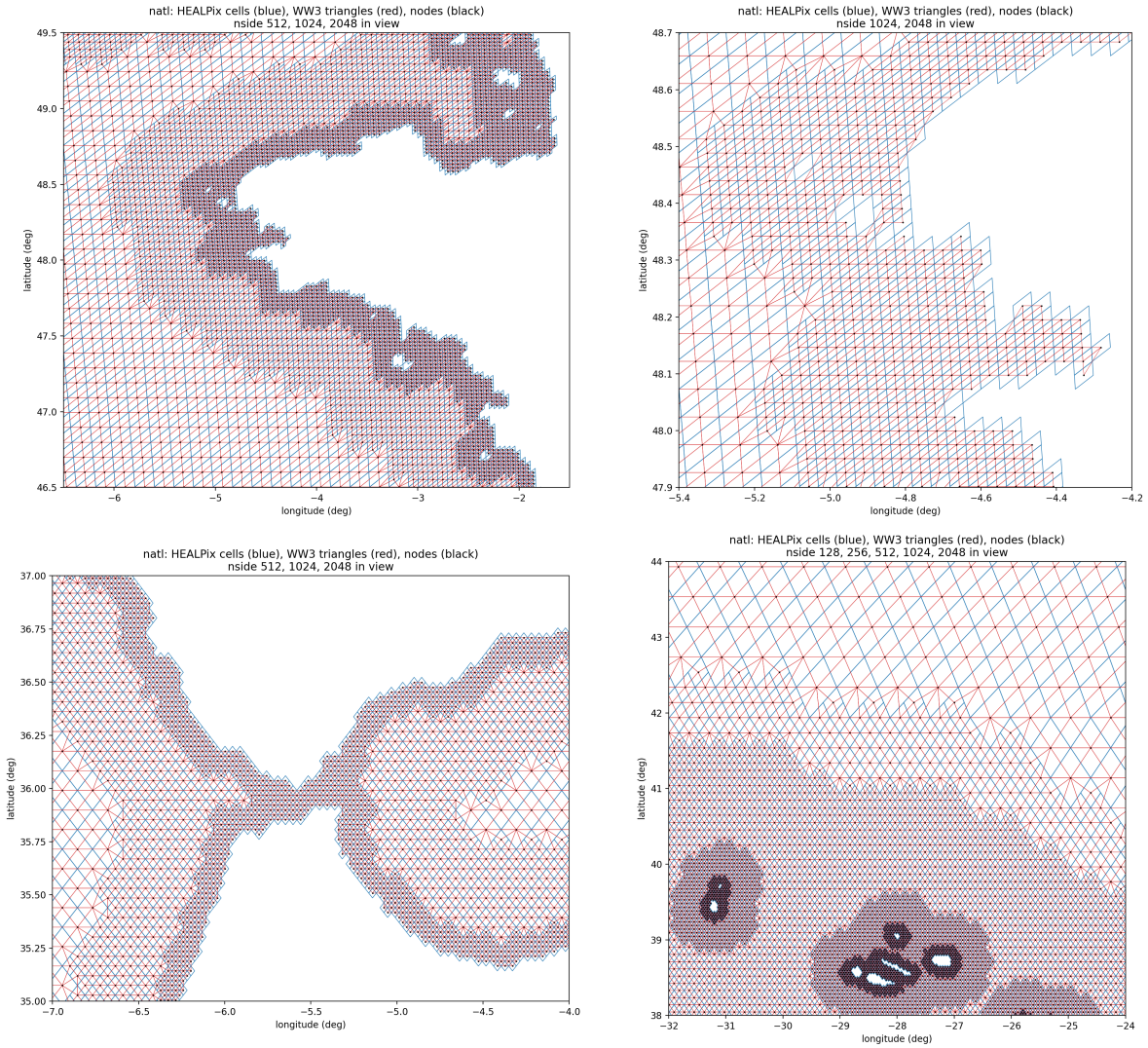


Figure 3: HEALPix cells (blue) and WW3 triangles (red) around Brittany, the Iroise Sea, the Strait of Gibraltar and the Azores (open ocean down to $N_{\text{side}} = 128$). Near 0° longitude above 41.8°N the cells are strongly sheared in longitude–latitude because they lie at the boundary of two polar base faces.

4.2 WW3 build and grid preprocessing

WW3 `develop` (commit `dea641c`) is built on Datarmor with Intel Fortran 19.1.3 and Intel MPI 2019.9 (`scripts/build_ww3.pbs`, 5.5 min), with the switches of `ww3/switch_hww3`: PDLIB with METIS domain decomposition, ST4, DIST/MPI, currents and wind input, IC0/IS0 (ice as an obstruction only). Two local dependencies are built because the system ones do not fit: netCDF-Fortran 4.6.1 (the system Fortran module was made by an old gfortran) on top of the system netCDF-C 4.8.1, and ParMETIS 4.0.3 with METIS 5.1.0. Intel's preprocessor drops the quotes of `-DENDIANNESS='big_endian'`, so `FILE_ENDIAN` is set in `constants.F90` before compiling.

The open-boundary file follows `READMSHOBC`: a comment line, then `index 0 node` lines, ended by a comment line; a bare node list is silently read as empty. Depths are written positive in the Gmsh file and turned negative by `UNST%SF = -1`. `ww3_grid` (PBS job 7893702, 35 s) then reads 502 690 sea points and 971 active boundary points and writes `mod_def.ww3` (260 MB).

4.3 First run: homogeneous wind

Before the forcing is available, `ww3_shel` is run for 12 h from a cold start (1 January 2026, 00–12 UTC) with a homogeneous 15 m s^{-1} westerly wind, no current, water level or ice, and no boundary spectra (`ww3/natl_test`, `scripts/natl_test.pbs`, PBS job 7893925). The run uses 56 MPI ranks on two Datarmor nodes, with the ParMETIS decomposition and the fully implicit scheme of Table 2 at $\Delta t = 600$ s.

The 72 time steps take 1337 s, i.e. 18.6 s per step and a speed-up of 32 over real time; the Jacobi–Gauss–Seidel solver stops after about 20 iterations per step, when 1.1 % of the nodes are not yet converged (threshold `JGS_PMIN` = 3 %). Memory use is 55 GB over the two nodes, the restart file is 2.3 GB and `ww3_ounf` writes the hourly fields in 5 s. A full month (4464 steps) would therefore cost about 23 h on 56 cores, or 1300 core-hours.

The wave field grows as expected (Fig. 4): the median H_s is 1.0 m after 3 h, 1.85 m after 6 h and 2.75 m after 12 h, with T_{02} rising from 2.6 to 4.8 s. H_s stays lower on the lee side of the land masses (east of Britain, Shetland, the Americas) and at the open boundaries, where no spectra are imposed. The highest values (3.9 m) are on the Norwegian shelf near 61°N, where the waves shoal. No NaN appears.

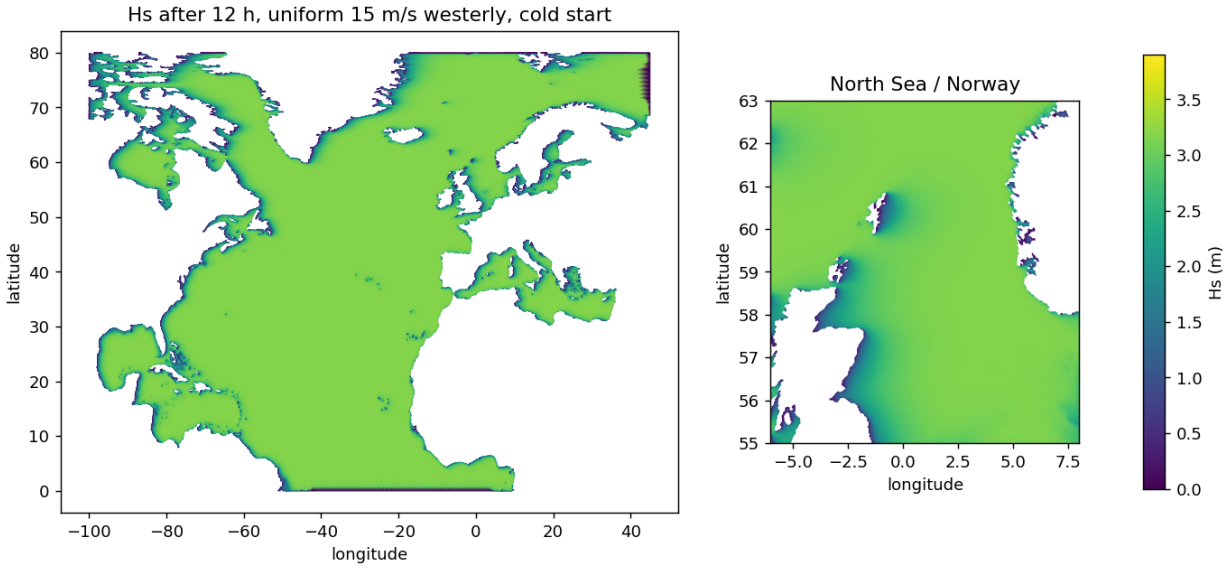


Figure 4: H_s after 12 h of a homogeneous 15 m s^{-1} westerly wind from rest (PBS job 7893925).

[TODO: forcing, boundary spectra, monthly run, validation against altimetry]

A Reproducing

- Table 1: `cd hww3-grid && cargo test -release island_multiscale - -nocapture`.
- Figure 1: `python scripts/plot_mesh.py /tmp/hww3_out_island tex/hww3/fig/island_mesh_zoom.p 41 49 39 44.5`.
- Table 2, Figures 2–3: `qsub scripts/natl_grid.pbs` on Datarmor, then `python scripts/plot_levels.py grids/natl/plot/nodes.npz fig/natl_levels.png` and `python scripts/plot_cells_tris.py grids/natl/plot/<zone>.npz fig/natl_<zone>.png`.
- Figure 4: `qsub scripts/natl_test.pbs` on Datarmor, then `python scripts/plot_hs.py runs/natl_test/ww3.202601.nc fig/natl_test_hs.png`.